RESEARCH ARTICLE

Measuring Post-Pandemic Road Traffic Recovery: Volume, Temporal Composition, and Observation-Network Sensitivity

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ABSTRACT
Longitudinal traffic observations show how urban mobility evolves after major disruptions. This paper presents a data-driven framework to assess post-pandemic road traffic recovery by distinguishing changes in volume, hourly composition, and observation-network coverage. We evaluate it using hourly motor-vehicle counts from Bologna, Italy, 2019–2025, with official recording-completeness records. We build a balanced panel of 70 detectors with stable identities, fully recorded days, and coverage in every monthly weekday/weekend stratum. With equal-month and fixed weekday/weekend weights, the volume index rises from 76.3 in 2020 to 94.1 in 2023, then falls to 89.3 in 2025 (2019 = 100). Pairwise panels confirm sensitivity to detector retention. By 2025, hourly composition is closer to its pre-pandemic distribution than volume is to its reference level, and the morning share is not persistently depressed. Recovery is incomplete and non-monotonic. The framework supports interpreting recovery through complementary measures while accounting for observation-network changes; uneven coverage does not support citywide generalisation or causal attribution.

KEYWORDS
COVID-19, traffic recovery, longitudinal traffic counts, pre-pandemic baseline, temporal composition, observation networks.

1. Introduction

Returning to pre-pandemic traffic levels is an intuitive description of recovery, but measuring it requires several distinctions. A rebound from the lockdown trough does not establish a return to the pre-pandemic level. Restored annual volume does not establish restored morning peaks, weekend activity, or traffic distribution across streets. Changes in the number of functioning detectors can also produce an apparent recovery without increased traffic at comparable locations.

Reviews document diverse transport responses to the pandemic (Benita, 2021; Abduljabbar, Liyanage, & Dia, 2022). For planning, it matters whether earlier patterns have returned or continue to change. Lower traffic volume can retain its previous daily rhythm, while an unchanged total can be distributed differently across hours. Neither observation alone identifies teleworking, modal shift, congestion, or economic recovery.

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Research has documented unequal recovery across modes, locations and periods of the day (Gao & Levinson, 2024; Z. Li, Wei, Zhang, Kong, & Ma, 2023; Zheng, Wang, Liu, Aloisi, & Zhao, 2024), while operational monitoring addresses reference periods and network continuity (Department for Transport, 2026; Cambridgeshire County Council, 2024). We extend this work by jointly evaluating all-day volume and hourly composition through 2025 against a full-year 2019 reference, and by assessing sensitivity to coverage and detector-panel membership. The contribution is integrated longitudinal evidence, not a new index or a causal explanation of pandemic effects.

The case study uses municipal hourly counts from Bologna, Italy, before, during, and after the acute pandemic disruption. The aim is to examine what an evolving urban detector archive can establish about recovery, rather than to claim an exceptional local trajectory or a citywide estimate.

Three questions organise the analysis. *RQ1*: how does calendar-standardised traffic volume evolve relative to 2019, and is the subsequent recovery monotonic? *RQ2*: how do weekday and weekend recovery differ, and does hourly composition recover in the same way as volume? *RQ3*: how sensitive are the conclusions to recording coverage, detector retention, and aggregation? These questions concern observed motor-vehicle passages, not unique journeys, passenger-car traffic alone, or citywide vehicle-kilometres.

The analysis combines a fixed seven-year panel with pairwise panels linking 2019 to each later year. All retained days must be fully recorded. Annual and monthly coverage requirements balance longitudinal comparability against detector retention, with selection based on observation availability rather than recovery outcomes. Alternative coverage thresholds, panel membership and aggregation weights provide sensitivity checks.

The rest of the paper is organised as follows. Section II reviews related work on recovery measurement. Section III introduces the case study and observation audit. Section IV defines the recovery measures and analysis design. Section V presents the results. Section VI discusses their interpretation and limitations, and Section VII concludes.

2. Related Work

2.1. Reference Levels and Calendar Alignment

Recovery measurement requires an outcome and a reference. A ratio to a pre-pandemic level measures the remaining gap; a post-trough slope measures rebound speed; a threshold crossing measures timing but may be temporary. These measures can therefore give different accounts of recovery.

Early count-based studies in Minneapolis–St. Paul (Liu & Stern, 2021) and multimodal monitoring in Santander (Aloi et al., 2020) quantify immediate disruption and initial adjustment; these short-run responses are not evidence of sustained restoration.

The British Department for Transport's road tracker uses approximately 275 automatic counters and compares each day with the corresponding weekday in the first week of February 2020 (Department for Transport, 2026). Its methodology warns that ordinary seasonal variation can reach approximately ±20%. Crossing 100 therefore need not mean matching the same season of 2019; different reference periods also complicate comparisons between modes.

Li et al. (Z. Li et al., 2023) study Chicago ride-hailing through March 2022 using ratios to 2019, holiday and weekday matching, smoothing, and trajectory clustering.

They capture spatially differentiated recovery trajectories, but observe a single service and impute missing tract information. Our analysis instead uses non-holiday monthly weekday/weekend strata without imputing counts. This simpler calendar matching does not control the mixture of individual weekdays within each group.

2.2. Recovery of Volume Versus Recovery of Peaks

Gao and Levinson (Gao & Levinson, 2024) analyse five-minute freeway data in six Californian cities, using working days in March–July 2018–2021. Their analysis of double-humped morning-peak geometry and differences between morning and evening recovery shows why daily volume alone is insufficient. Traffic-flow recovery must also be distinguished from travel-time recovery.

Li, Zhao and Wang's Glasgow dataset spans October 2019–September 2023 and shows a reappearing but incompletely restored morning peak (Y. Li, Zhao, & Wang, 2025). Its SCOOT units and aggregation differ from our vehicle counts, and its 2019 reference covers only part of the year. Aletta et al. (Aletta, Gemma, Mannini, & Patella, 2022) compare Rome's morning peak in October 2019 and October 2022 using probe data, demand estimation and assignment. They find approximately 10% lower modelled vehicle-kilometres. These studies provide temporal comparisons, but their percentages are not directly comparable with all-day recovery indices.

Dutch mobility-panel evidence likewise distinguishes returning activity from changed timing, motives and modes (van der Drift, Wismans, & Olde Kalter, 2022). We separate hourly levels from shares, without claiming subhourly peak geometry or identified trip purposes.

At the individual level, Ferrari and Mamei (Ferrari & Mamei, 2011) use topic models to identify recurrent routines from Google Latitude mobility records. Although their study does not address pandemic recovery, it provides relevant methodological context for distinguishing recurring mobility patterns from aggregate activity levels. Our hourly profiles describe the timing of detector passages, not individual routines or the purposes behind them.

2.3. Changing Observation Networks and Uneven Recovery

Glasgow's refinement retains 470 of 1,033 sensors after spatial, temporal, interpolation and zero-series checks (Y. Li et al., 2025). Cambridgeshire's June 2024 monitoring retains fourteen uninterrupted strategic-road sensors and warns that changing membership makes sums incomparable across reports (Cambridgeshire County Council, 2024). A recovery percentage therefore belongs to a declared observation network, not automatically to a city.

Spatially differentiated trajectories also appear in Chicago (Z. Li et al., 2023) and, at a different scale, Chinese provinces (Zhang, Zhao, Peng, & Gong, 2022). Detector differences can additionally reflect rerouting or instrument changes; they are not automatically behavioural effects.

Glasgow's associated analytical study examines traffic in relation to the built environment and spatial dependence (Y. Li, Zhao, & Wang, 2024). Our focus is instead the robustness of longitudinal recovery summaries. Other studies distinguish recovery of frequency, distance and activity space in Kunming transit records (Wang, Huang, Yang, & Levinson, 2022), and heterogeneous observed and intended train-travel trajectories in Dutch panels (Kroesen, Le, De Vos, Ton, & de Bruyn, 2023). These service-specific

outcomes and intentions do not directly measure general road traffic.

The relationship between observation source and mobility target also arises in Mamei et al. (Mamei, Bicocchi, Lippi, Mariani, & Zambonelli, 2019), who evaluate OD matrices derived from mobile phone call detail records, including population scaling and road-network projection. Their work concerns mobility inference rather than post-pandemic recovery, but offers a related perspective on evaluating what source-specific observations can establish. Here, the corresponding boundary is between recovery of passages at a declared detector panel and recovery of citywide journeys or travel demand.

[Table 1 about here.]

2.4. Behavioural Mechanisms and Scope of Inference

Zheng et al. (Zheng et al., 2024) relate remote work to US vehicle-miles and transit ridership, finding different relationships by mode. Nello-Deakin et al. (Nello-Deakin, Diaz, Roig-Costa, Miralles-Guasch, & Marquet, 2024) identify normalised teleworking but limited broader mobility-regime transformation in metropolitan Barcelona. These findings suggest possible mechanisms, not coefficients transferable to our detector panel.

Beijing's longitudinal mode-switch analysis (Gao & Zhao, 2024) uses movement evidence beyond fixed counts. Without observations of workplace attendance, trip purposes or mode choices, our traffic counts cannot identify teleworking effects or support comparable behavioural attribution.

2.5. Positioning and Contribution

Together, these studies motivate examining whether recovery persists across time, outcomes and observation designs. We address this question by combining all-day volume and hourly shares through 2025 with comparisons between a fixed seven-year panel, alternative-coverage panels and pairwise panels. Normalised indices, temporal-profile analysis and panel cleaning are established tools; the added evidence concerns their joint application to the persistence and robustness of observed recovery, not their invention or a universal urban trajectory.

3. Case Study and Observation Audit

3.1. Source and Measurement Unit

Seven municipal CSV files obtained from the Municipality of Bologna's open-data service[1] contain daily records with twenty-four hourly count fields, one file for each year from 2019 to 2025 (Comune di Bologna, 2026). Each spans January 1 to December 31 and includes detector codes, station identifiers, positions, device types, status, and street information. Counts measure motor-vehicle detections at a detector. They do not distinguish passenger cars, identify occupants, or show whether a vehicle passes another detector. Summing counts across detectors therefore measures observed passages, not unique urban journeys.

[1]Municipality of Bologna open-data portal: `https://opendata.comune.bologna.it/`.

The analytical area is a fixed observation box within latitude 44.40–44.60 and longitude 11.20–11.50; it does not follow the municipal boundary. We retain active loop detectors with valid station identities, coordinates and dates, and finite nonnegative hourly counts. Table 2 summarises the annual source audit; the numerical outputs retain the full audit.

[Table 2 about here.]

3.2. Identity, Aliases and Recording Completeness

Municipal documentation warns that a detector's operational code can change. We therefore track stations over time using the municipality's stable station reference. Official recording-status records are linked before duplicate daily entries are reconciled; their provenance is documented in the accompanying source manifest. Identical copies of a station's daily counts are retained once, not added. If their recording status differs, a deterministic rule retains the most complete copy. Conflicting count copies are excluded, as are conflicting recording-status entries that cannot certify complete measurement.

The official description distinguishes full, absent and partial recording. We require exactly 100% recording for every hour of a retained day. We neither scale up incomplete days nor treat absent records as zero traffic. Requiring complete days gives hourly profiles and daily totals the same observation exposure, at the cost of substantial coverage. It is stricter than retaining individual complete hours and is not the only defensible use of the archive.

We remove the two daylight-saving transition dates in each year because the fixed twenty-four-bin exports cannot resolve repeated or missing clock hours. We also exclude a station from longitudinal analysis if any retained candidate position lies more than 100 m from its pooled median position. This rule removes six station identities across the seven annual snapshots, but does not independently verify that every remaining detector measured the same lane or approach throughout the period.

Observation availability changes substantially: the audit records approximately 150,000 complete detector-days in 2019, 169,000 in 2022, and 100,000 in 2024. These count detector-days, not distinct dates. Raw archive sums would therefore combine traffic changes with changes in observation exposure and network membership. Full-recording flags also do not independently establish detector accuracy. The wider archive includes all-zero days with full-recording flags; we retain them for transparent diagnostics rather than assume they are missing observations.

3.3. The Balanced Panel and Its Geography

The primary balanced panel requires at least 120 complete dates in *each* year. Each detector/year/month must also contain at least two non-holiday weekdays and one non-holiday weekend day, all fully recorded. Holidays include Italian national public holidays, Easter and Easter Monday, and Bologna's patronal day. These coverage-based criteria do not select detectors according to their recovery outcomes. The target is non-holiday traffic under a fixed calendar mixture, not actual annual totals including holidays.

These rules retain 70 detectors in 24 occupied 1 km blocks, with 119,449 complete detector-days before holiday exclusion, including 288 zero-total days. Figure 1 compares the retained positions with the 907 distinct source stations observed in the

analytical box during 2019–2025 after basic detector and position checks. The panel includes western and eastern corridors, but its uneven coverage cannot be assumed representative of the city. Requiring 240 complete days in every year, together with the monthly requirements, leaves no qualifying seven-year panel.

[Figure 1 about here.]

4. Recovery Measures and Analysis Design

4.1. A Fixed Calendar and Detector Panel

Let $\bar{q}_{symk}$ denote mean daily volume over complete non-holiday days at detector s, in year y, month m, and day type k, with $k = 0$ for weekdays and $k = 1$ for weekends. Define the standardised monthly and annual volumes as

$$v_{sym} = \frac{5}{7}\bar{q}_{sym0} + \frac{2}{7}\bar{q}_{sym1}, \qquad v_{sy} = \frac{1}{12}\sum_{m=1}^{12} v_{sym}. \tag{1}$$

Weights remain fixed across years. Equal-month weighting prevents differences in observed-day counts from changing the seasonal mixture. The $5/7, 2/7$ weights similarly fix the weekday/weekend mixture. We do not impute missing strata: the primary panel requires observations in all strata.

For fixed panel S, the principal volume index is

$$I_y = 100\frac{\sum_{s\in S} v_{sy}}{\sum_{s\in S} v_{s,2019}}. \tag{2}$$

Monthly indices compare each month's standardised sum with the corresponding month of 2019. The annual index is a ratio of standardised sums, not an unweighted average of monthly ratios. Higher-volume detectors contribute more than lower-volume detectors; none is weighted by a known sampling probability. To describe detector-level variation, we also report the median of $100v_{sy}/v_{s,2019}$ and the shares reaching 95 and 100. Detectors with a zero 2019 denominator are ineligible because their individual ratios are undefined; later-year zeros, if present, would be retained.

The threshold of 95 is a descriptive near-reference convention, not a statistical equivalence margin. Three consecutive monthly indices at or above a threshold provide a persistence diagnostic, rather than an irreversible recovery date. The reference is observed 2019, not a modelled no-pandemic counterfactual. With only one complete pre-pandemic year, we cannot estimate the secular pre-pandemic trend or the traffic level that would have occurred without COVID-19.

4.2. Day Types and Temporal Composition

Weekday and weekend indices are calculated separately using equal-month averages over the same detector panel. Hourly means use the same complete dates as daily totals and are then averaged equally across months. For day type k, let Q_{ykh} be the sum of these standardised hourly means across panel detectors at hour h in year y. We

report both hourly levels and hourly shares,

$$p_{ykh} = \frac{Q_{ykh}}{\sum_{j=0}^{23} Q_{ykj}}, \tag{3}$$

which separate the distribution across hours from total volume. A small compositional change can therefore coexist with a large volume deficit. We compare each year's shares with 2019 using Jensen–Shannon distance with base-two logarithms and total variation, $\frac{1}{2}\sum_h |p_{ykh} - p_{2019,kh}|$. Both range from zero to one and describe distributional differences; they do not test behavioural equivalence.

Four broad weekday windows provide interpretable contrasts: morning 07–09, evening 17–19, midday 10–16, and night 00–06. Window indices are ratios of summed hourly levels to their 2019 values; window shares use the full day's total as the denominator. These clock-time windows do not identify commuting, leisure, or freight traffic and do not partition the full day. Their definitions were not selected according to observed recovery results.

4.3. Spatial Resampling and Temporal Trends

We resample fixed 1 km spatial blocks with replacement 2,000 times, retaining all detectors in each sampled block and recalculating the ratio of sums. The 2.5th and 97.5th percentiles describe sensitivity to the representation of observed blocks. The network contains 24 occupied blocks, is geographically selective, and has dependence between adjacent blocks. These intervals do not quantify citywide uncertainty, recording error, or within-stratum day-to-day sampling variation. The 2019 reference is held fixed within each resampled block.

To summarise the later trajectory separately, we regress the monthly index on calendar-month indicators and a linear time term over 2022–2025. Newey–West covariance (Newey & West, 1987) with twelve lags allows for serial dependence. A second fit over 2023–2025 assesses sensitivity to the starting year. This alternative window does not identify a structural break or an independently validated change point. The respective samples of forty-eight and thirty-six monthly observations provide limited information for asymptotic interval estimation.

4.4. Observation-Design Sensitivity

We assess coverage sensitivity by changing the minimum annual number of complete days to ninety, 180 and 240 while retaining the monthly requirements. A positive-day diagnostic excludes zero days and recomputes eligibility and indices, making the effects of this potentially selective choice explicit. An equal-block summary pools volumes within each block, forms block-specific ratios to 2019, and averages those ratios equally. It changes geographical weighting without adding locations.

Separate pairwise panels apply the same 120-day annual and monthly-stratum criteria to 2019 and one later year, without requiring retention in intervening years. These panels broaden spatial coverage but change membership between annual comparisons, so they do not describe a continuous trajectory for a common network. Comparing them with the fixed panel assesses part of the sensitivity to detector retention; agreement does not establish that either network represents all urban traffic.

5. Results

5.1. A Strong Initial Rebound Without Complete Restoration

The fixed panel shows a large initial disruption followed by a substantial but incomplete rebound (Table 3; Fig. 2). The annual index is 76.3 in 2020, 86.7 in 2021 and 92.4 in 2022 (2019 = 100). The lowest monthly index is 27.7 in April 2020 relative to April 2019, showing that the annual average conceals a much deeper temporary trough.

[Table 3 about here.]

[Figure 2 about here.]

The annual index reaches 94.1 in 2023, then declines to 92.6 in 2024 and 89.3 in 2025. Recovery is therefore not a monotonic return to 100. The 2025 volume deficit is approximately 10.7% of the standardised 2019 level. The spatial-resampling interval for the 2025 index is approximately 84.9–93.8; it does not extend inference beyond the observed corridors.

After the initial trough, the first sequence of three consecutive monthly indices at or above 90 begins in June 2021. No such sequence reaches 95 or 100 through the end of 2025. Under this descriptive threshold criterion, the substantial rebound does not amount to sustained near-reference volume.

The month-adjusted slope over 2022–2025 is -1.11 index points per year, with a heteroskedasticity- and autocorrelation-consistent (HAC) interval of $[-2.19, -0.02]$. Over 2023–2025, the slope is more negative: -2.47 points per year, with interval $[-3.05, -1.89]$. Both fits summarise a later decline rather than continued recovery towards 100. Their difference shows sensitivity to the time window; neither identifies an enduring trend, a pandemic legacy, or a policy effect.

5.2. Volume and Temporal Structure Recover Differently

Weekday and weekend indices differ on the same panel (Fig. 3). In 2021, the weekday index is 88.6 and the weekend index is 80.6. By 2025, the gap is smaller, at 89.8 and 88.0, respectively. Weekend traffic in this panel does not recover earlier or more completely than weekday traffic; this finding should not be generalised to cities or mobility services where the reverse pattern has been observed.

[Figure 3 about here.]

In 2025, the weekday morning-window index is 90.3, compared with 88.3 in the evening, 91.8 at midday, and 88.9 during 00–06 (Table 4). Morning volume remains below its 2019 level, but its share of weekday detections is slightly higher, by 0.09 percentage points. The evening share falls by 0.23 percentage points, while the midday share rises by 0.82 percentage points. Thus, lower volume coexists with substantial retention of the 2019 within-day pattern (Fig. 4). The morning-volume deficit does not, in this case, indicate a persistently reduced morning share.

[Table 4 about here.]

[Figure 4 about here.]

The aggregate hourly-share distances from 2019 also decrease. Weekday Jensen–Shannon distance is approximately 0.050 in 2021 and 0.018 in 2025; the corresponding

weekend distances are approximately 0.066 and 0.016. Total variation in 2025 is approximately 0.017 on weekdays and 0.015 at weekends. These small compositional distances coexist with volume indices well below 100, indicating that broad temporal composition is closer to its reference than volume. Aggregation may nevertheless conceal offsetting detector-level changes, and hourly bins cannot reveal changes in sharper subhourly peaks.

5.3. Spatial Differences and Network Sensitivity

The median detector ratio in 2025 is 89.1, close to the ratio-of-sums index, with an interquartile range of approximately 82.1–99.8. Only 30.0% of detectors reach 95% of their standardised 2019 level, and 25.7% reach 100. Some locations exceed the reference, while most remain below it. Figure 5 shows this variation within the observed network, not a citywide demand distribution.

[Figure 5 about here.]

The coverage checks yield similar 2025 indices despite differences in panel size (Table 5). Requiring ninety complete days per year retains 72 detectors and gives an index of 89.2, compared with 89.3 for the primary panel. Requiring 180 days retains 42 detectors and gives 90.8; no panel satisfies the 240-day requirement. Excluding zero days and reapplying eligibility retains 67 primary-panel detectors and gives 89.2. This diagnostic changes both day selection and panel membership. It neither identifies individual zero observations as detector failures nor justifies their default exclusion.

[Table 5 about here.]

Pairwise panels retain 196–330 detectors, compared with 70 in the seven-year panel. The 2019–2025 pair retains 224 detectors and gives an index of 88.6, also indicating incomplete restoration. This broader comparison does not contradict the fixed panel's endpoint finding, but does not remove geographical selection. Because membership differs across pairwise panels, differences between their annual indices cannot be interpreted as within-network growth.

Equal weighting of the 24 fixed-panel blocks gives a 2025 index of 88.6, compared with the primary index of 89.3. Figure 6 brings the alternative designs together while distinguishing changes in weighting from changes in panel membership.

[Figure 6 about here.]

By contrast, aggregation without fixed detectors or calendar exposure gives a substantially different account. Summing all complete detector-days would produce indices of approximately 107.5 in 2022 and 63.4 in 2024, compared with 92.4 and 92.6 in the primary panel. These archive-wide values do not establish a traffic overshoot followed by a comparable collapse, because both the set and number of measurements change. Dividing by available detector-days reduces exposure differences but still allows the mix of roads, dates and traffic levels to change. Neither shortcut replaces an explicitly defined longitudinal target.

6. Discussion

6.1. Interpreting an Incomplete and Non-Monotonic Recovery

The observed sequence combines a rapid rebound from the acute disruption with a later period below the 2019 level. This does not imply an equivalent deficit in urban activity: the counts measure passages at selected detectors, not economic activity or comprehensive travel behaviour. Nor is a return to 100 necessarily a desirable transport-policy outcome. Fewer motor-vehicle movements could accompany improved accessibility, reduced travel needs, or displacement to unobserved roads; these data cannot distinguish those possibilities.

Californian freeway and Glasgow observations motivate close attention to morning peaks (Gao & Levinson, 2024; Y. Li et al., 2025). In our selected corridors, the morning share is not lower in 2025 despite lower morning volume. This distinction between recovery of absolute volume and hourly composition does not refute teleworking effects elsewhere. Remote work, school travel, delivery activity, road access and rerouting are possible contributors to the observed patterns, but the detector data cannot separate their effects.

Rome's peak-period vehicle-kilometre comparison (Aletta et al., 2022) further illustrates the limits of cross-study interpretation. Similar percentage deficits in a single window would not establish similar changes in annual travel, distance, emissions or congestion. Our counts provide no travel speeds, occupancy, route lengths or vehicle technology. Estimating emissions or delay would require additional observations and an explicit model.

6.2. Limits of the Evidence

Three limitations frame these findings. First, the geographically uneven seven-year panel cannot be assumed representative of city traffic. Pairwise and coverage checks assess sensitivity within the archive, but cannot establish citywide representativeness. Second, full-recording flags and position checks do not independently verify detector accuracy or unchanged lane coverage. Retaining zero-total days avoids assuming that they are failures, while the positive-day diagnostic shows the sensitivity to excluding them. Third, the single pre-pandemic year supplies an observed reference, not an estimated pre-pandemic trend or no-pandemic counterfactual. Spatial-resampling intervals and short later-period regressions do not remove these limits or support causal attribution.

6.3. What the Case Contributes Beyond Its Location

The transferable contribution is a reporting discipline, not a universal recovery rate: define whether recovery concerns a level, a slope, a sustained threshold crossing, or distributional similarity; specify the reference calendar and exposure weights; distinguish traffic changes from observation-network changes; and report location-level variation alongside pooled summaries. These principles extend the practical lessons of operational monitoring (Department for Transport, 2026; Cambridgeshire County Council, 2024) and longitudinal sensor refinement (Y. Li et al., 2025).

Normalising an index to 100 removes units but can hide incompatible reference periods and outcomes. A full-year 2019 reference, a February 2020 week, a modelled October peak, smartphone visits and ride-hailing use define different comparisons. The

literature therefore informs expectations about multidimensional and uneven recovery, rather than providing a ranking of directly comparable percentages.

7. Conclusion

This study examined post-pandemic traffic recovery through volume relative to 2019, temporal composition, and observation-network comparability.

RQ1—Volume and trajectory. The fixed-panel calendar-standardised index rises from 76.3 in 2020 to 94.1 in 2023, then declines to 89.3 in 2025. This is incomplete and non-monotonic recovery of observed passages relative to 2019, not an identified long-term pandemic effect.

RQ2—Temporal composition. Weekday and weekend recovery differ. By 2025, aggregate hourly shares are much closer to their 2019 distributions than volume is to its reference level. The morning share is slightly higher, not persistently depressed. Similarity of the daily pattern and recovery of volume are therefore separate findings.

RQ3—Observation sensitivity. Requiring fully recorded days, at least 120 complete dates per year, and observations in every monthly day-type stratum retains 70 detectors. Broader pairwise comparisons also indicate incomplete restoration, but do not provide a representative citywide estimate. Aggregation without fixed detectors or calendar exposure can imply a substantially different trajectory.

The Bologna case shows why recovery estimates should be reported with their target, reference period, calendar weights, retained network and uncertainty limits. Longitudinal traffic archives can inform post-pandemic comparisons only if changes in the measurements are distinguished from changes in the traffic being measured.

Data and Code Availability

The municipal count and recording-status sources are available under CC BY 4.0; the supplied count snapshots remain unchanged. The accompanying code freezes annual source metadata and completeness exports, audits detector-days, generates all numerical tables and figures, and tests audit reconciliation and index definitions. The reproducibility manifest records exact source URLs, hashes, software versions and derived-output hashes.

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Disclosure statement

The authors report no competing interests.

Author contributions

Nicola Bicocchi: conceptualisation and formal analysis. Marcello Pietri: software. Marco Mamei: writing—original draft and writing—review and editing. All authors approved the final manuscript and agree to be accountable for all aspects of the work.

Ethics statement

Ethical approval was not required for this study.

Declaration of generative AI use

Generative AI was used only for proofreading and preparing submission materials, not for data analysis or interpretation.

Data availability statement

The data and code supporting this study will be available after publication upon request to the corresponding author.

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Biographical notes

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Table 1. Selected recovery-measurement approaches. Targeted methodological comparison, not a systematic review.

Study	Recovery measurement	Main comparability issue
DfT (Department for Transport, 2026)	Same-weekday reference-week index	Seasonal variation remains.
Chicago (Z. Li et al., 2023)	Calendar-matched ratios and trajectory clusters	Ride-hailing is a selective service.
California (Gao & Levinson, 2024)	Five-minute peak geometry	Shape is distinct from volume.
Rome (Aletta et al., 2022)	October peak-period VKT	Modelled distance is not passages.
Glasgow (Y. Li et al., 2025)	Refined network and hourly distributions	Continuity and zero-series selection.
Cambridgeshire (Cambridgeshire County Council, 2024)	Matched-period sensor panels	Membership defines the comparison.

Table 2. Source audit before cross-year stability checks and panel selection. Complete detector-days require all 24 hourly recording flags to equal 100%. Zero-total days are retained, not treated as missing.

Year	Raw rows	Complete days	Detectors	Zero days
2019	287,747	149,564	771	1,595
2020	284,787	144,487	785	683
2021	290,531	160,804	790	1,080
2022	302,872	168,955	874	1,590
2023	313,067	133,403	849	2,170
2024	309,561	99,559	826	1,252
2025	304,824	113,976	781	2,358

Table 3. Balanced-panel annual recovery. The index is a ratio of standardised sums (2019 = 100); intervals use spatial-block resampling. The median and threshold share describe individual detectors, not independent samples of the city.

Year	Index	95% interval	Median	≥ 95 (%)
2019	100.0	[100.0, 100.0]	100.0	100.0
2020	76.3	[74.5, 77.9]	76.8	4.3
2021	86.7	[84.3, 88.9]	85.6	17.1
2022	92.4	[89.6, 95.5]	90.9	22.9
2023	94.1	[90.9, 97.3]	92.5	32.9
2024	92.6	[89.1, 96.5]	91.3	35.7
2025	89.3	[84.9, 93.8]	89.1	30.0

Table 4. Weekday window indices, 2019 = 100. [a]Change in the window's share of all weekday detections, 2025 minus 2019, in percentage points. Windows do not partition the full day.

Window	2020	2022	2025	Δ share[a]
Morning 07–09	78.9	95.4	90.3	+0.09
Evening 17–19	79.2	94.1	88.3	-0.23
Midday 10–16	82.3	94.5	91.8	+0.82
Night 00–06	65.5	87.6	88.9	-0.03

Table 5. Observation-design sensitivity. All indices use 2019 = 100. S_{2025} is the detector count for the 2025 comparison. Pairwise membership changes across years. A dash indicates no qualifying panel, not zero traffic.

Design	S_{2025}	2020	2023	2025
Balanced 90 days	72	76.3	94.2	89.2
Balanced 120 days	70	76.3	94.1	89.3
Balanced 180 days	42	76.9	93.4	90.8
Balanced 240 days	0	–	–	–
Positive-day diagnostic	67	76.2	94.1	89.2
Pairwise balanced	224	76.7	94.3	88.6
Equal spatial blocks	70	76.4	93.7	88.6

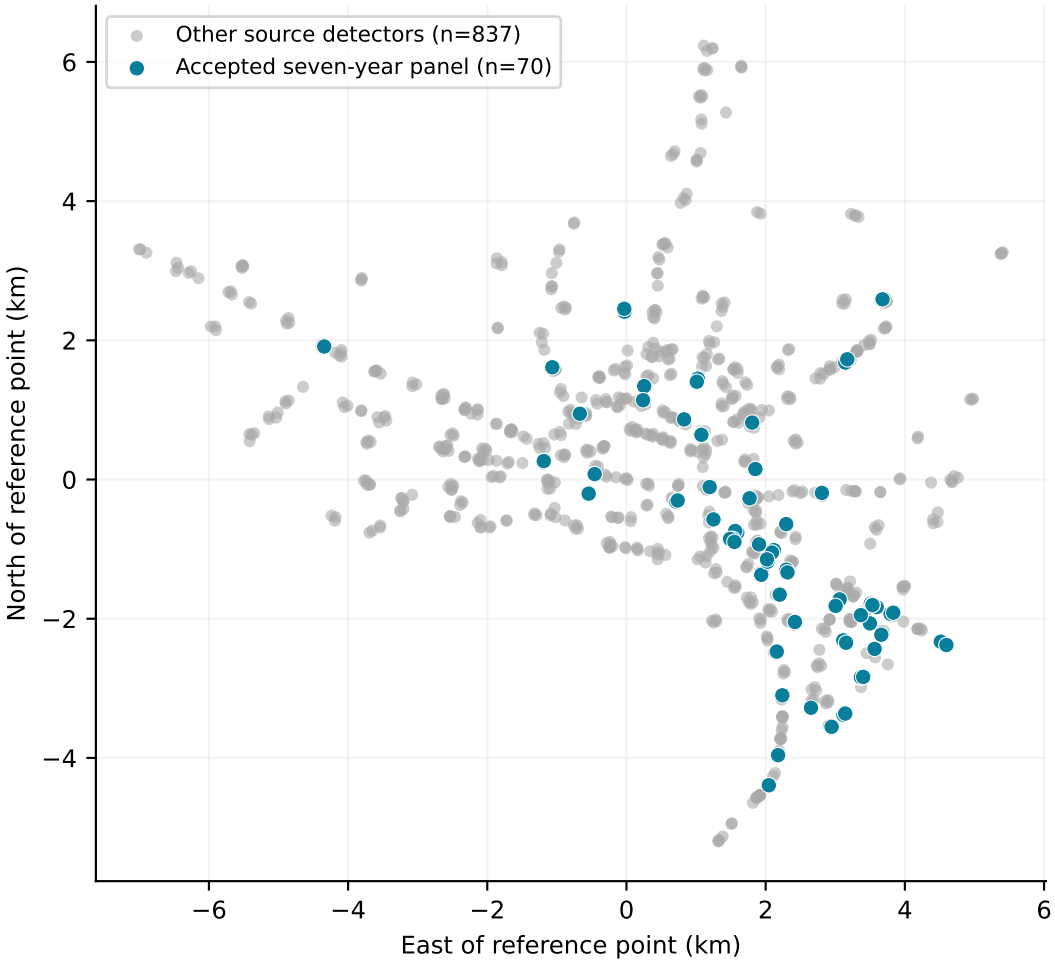

Other source detectors (n=837)
Accepted seven-year panel (n=70)
North of reference point (km)
East of reference point (km)
6
4
2
0
−2
−4
−6
−4
−2
0
2
4
6

Figure 1.

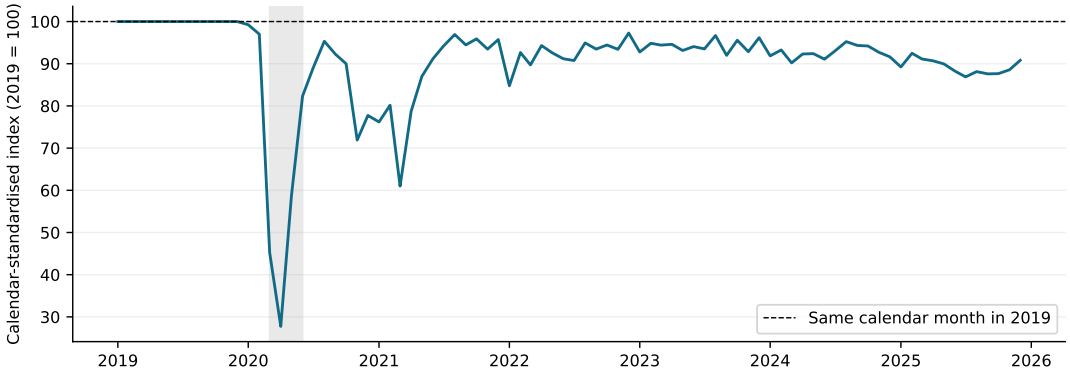

Calendar-standardised index (2019 = 100)
100
90
80
70
60
50
40
30
2019
2020
2021
2022
2023
2024
2025
2026
Same calendar month in 2019

Figure 2.

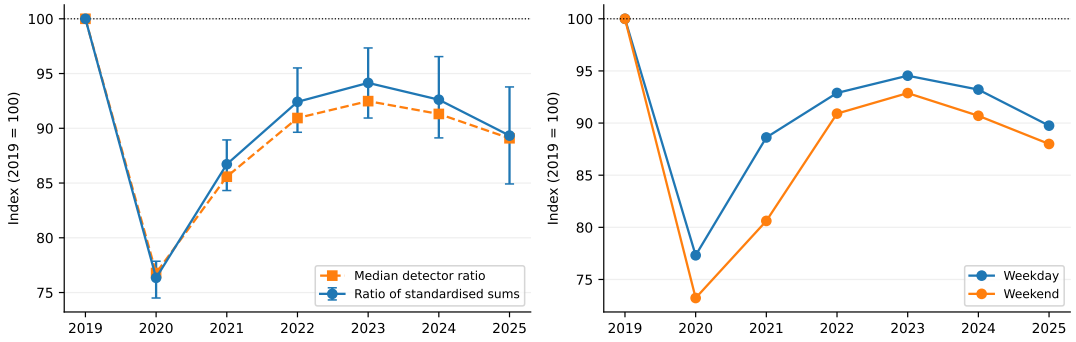

Index (2019 = 100)
100
95
90
85
80
75
2019
2020
2021
2022
2023
2024
2025
Median detector ratio
Ratio of standardised sums
Index (2019 = 100)
Weekday
Weekend

Figure 3.

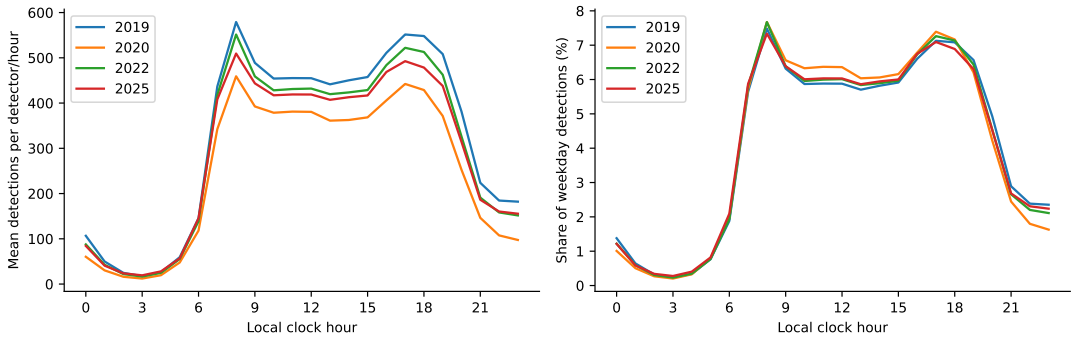

2019
2020
2022
2025
Mean detections per detector/hour
Local clock hour
Share of weekday detections (%)

Figure 4.

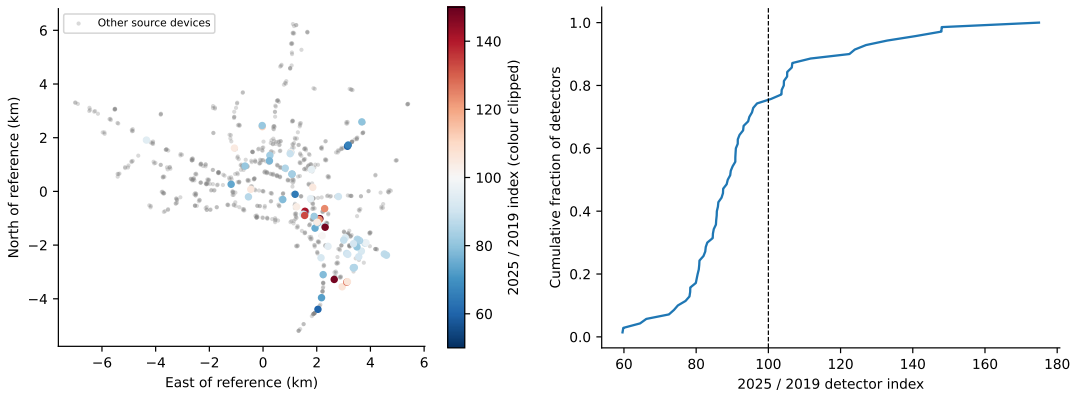

Other source devices
North of reference (km)
East of reference (km)
2025 / 2019 index (colour clipped)
Cumulative fraction of detectors
2025 / 2019 detector index

Figure 5.

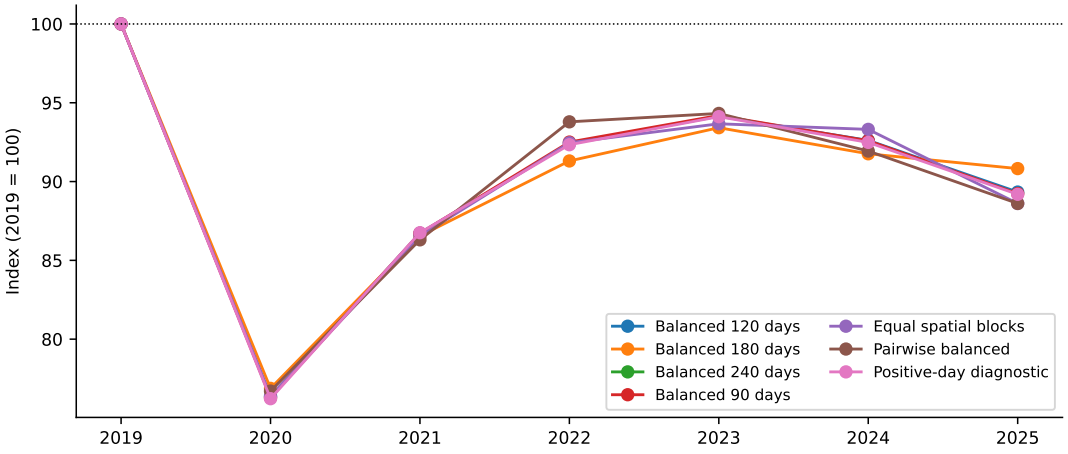

100
95
90
85
80
Index (2019 = 100)
2019
2020
2021
2022
2023
2024
2025
Balanced 120 days
Balanced 180 days
Balanced 240 days
Balanced 90 days
Equal spatial blocks
Pairwise balanced
Positive-day diagnostic

Figure 6.

Figure captions

Figure 1. Accepted seven-year detectors (blue, 70 stations) relative to all other source stations (grey, 837) in the analytical box. The overall network is the union of station positions in the 2019–2025 source files after basic active-loop, identity and position checks, not a claim that all stations operated simultaneously. Counts refer to distinct stations; nearby symbols may overlap.

Figure 2. Monthly volume relative to the same month of 2019, using the fixed detector panel and fixed weekday/weekend weights. Shading marks March–May 2020 as a visual reference, not an estimated intervention boundary. Values describe observed corridors, not total city travel.

Figure 3. Left: annual ratio-of-sums index and median detector ratio. Error bars are spatial-block percentile intervals, not population confidence bounds. Right: separately standardised weekday and weekend indices on the same detector panel.

Figure 4. Weekday hourly levels (left) and normalised shares (right) for selected years, with equal-month averaging. For each detector, all hours use the same fully recorded days. Similar shares do not imply restored volume or identify trip purposes.

Figure 5. Left: fixed-panel positions coloured by the 2025/2019 detector index; grey dots show the wider source network. Colour limits are clipped at 50 and 150 only for display. Right: empirical distribution of uncapped detector indices. Uneven spatial coverage limits geographical interpretation.

Figure 6. Alternative coverage and weighting designs. Lines joining pairwise results are visual guides across different cohorts, not one longitudinal network. The 240-day design has no eligible sites and therefore no plotted trajectory.